$$\hat{U}(t) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^n}{n!} \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \cdots \\ \times \int_0^{t_{n-1}} \mathrm{d}t_n \, \mathcal{T}_{\leftarrow} \left\{ \hat{H}_{\mathrm{int}}(t_1) \hat{H}_{\mathrm{int}}(t_2) \cdots \hat{H}_{\mathrm{int}}(t_n) \right\}. \quad (5.96)$$